

Manual

Inspection Result Documentation SMA-PED 8.2

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1 Inspection Result Documentation

Possible fields of application

This function package is used if inspection data are collected on a mobile device. Also, measured values for variable characteristics or number of defect parts for attributive characteristics can be collected. The function package supports the inspection by displaying linked inspection documents.

Implementation notes

A typical application for the package is if large, heavy parts to be inspected cannot be collected in a central inspection station. In this case the "mobile inspection station" travels to the parts to be inspected.

In order to use this package inspection points must be generated beforehand. The generation of inspection points requires in turn an auditable inspection step.

Inspection points can be manually generated via the MOC or the AIP. Also, defined events in the production environment can automatically generate inspection points. However, it is required that an order or an inspection step is logged on in the AIP. Possible events are the following:

- Completing a piece or time interval
- Order logon
- Shift change
- Output batch change
- Machine status change

Integration

Documented inspection results using this function package on a mobile device is directly related to the function of the inspection requirement, inspection step and inspection point in the MOC.

Functions Scope

What does this function package offer, especially relating to the documented functions of the license?

This package offers the following performance features to document inspection results on a mobile device:

- Guided inspection result collection displaying quality-relevant information (i.e. specified limits).
- Collecting measured values for variable characteristics and the number of defect part for attributive inspections.
- Completion of inspection points with automatically generated inspection point usage decision.
- Visualization of inspection documents (drawings, inspection instruction etc.) which have been assigned to the corresponding inspection plan.

2 List of Inspection Points

On the left hand side a list containing all auditable inspection points appear. The list can be restricted using a filter. A match code filter is available for the user. Content to be filtered in a text is replaced by the symbol "*". Using the symbol "?" an individual symbol can be replaced. The complete content of the inspection point list can be filtered. The following information is part of the inspection point list:

- Order - operation
Both are separated with a hyphen and are shown in the first line of the inspection point. Example of a display format: 12345678 – 0010“ („12345678“ is the order number, „0010“ is the operation).
- Article number - article description - date of when inspection point was created
The information is part of the 2. line and are also separated by a hyphen.

Also, the status of an inspection point result is highlighted by a colored LED symbol. The following color assignments are valid:

- o.k. => green LED
- fail => red LED => rote LED

A special filter containing can be requested by the filter symbol to look for the area and the inspection point status. This is in addition to the possibility to filter using a match code. To facilitate the filter process corresponding selection lists are available. .

In the right display area detailed information is shown in the list for the selected inspection point. The parallel display of the left and right display area is dependent on the selected device. On a smartphone only the inspection list is shown. In order to show details the required inspection point should be selected.

The detailed display enables using special function buttons for the following create/process function:

- Request the characteristics list
- Editing inspection point
- Completion of inspection point

Based on the inspection result of the corresponding inspection point characteristics when the function "Completion of inspection point" is triggered, the inspection point usage decision is automatically identified and the inspection is completed at the same time.

3 Process Inspection Point

The dialog to process inspection point details is separated into groups:

- Order
- Identification
- Additional fields

The field content of the group "Order" contain information about the subordinate order and cannot be edited. Amongst others order and operation number and information about the article are shown.

Referring to the group "Identification" a clearly defined inspection point number and inspection result based on previous inspections is shown. The field content is again not editable.

All other inspection point fields can be edited. Some of these field only carry a general description "Field 1" up to "Field 7". This reasons for it is that in the CAQ system option these fields contain different labels per area in the MOC presentation. Currently the area dependent labelling is not supported by the SMA.

4 List of Characteristics

On the left hand side a list containing all characteristics for previously selected inspection points appears. The list can be restricted using a filter. A match code filter is available for the user. Content to be filtered in a text is replaced by the symbol "*". Using the symbol "?" an individual symbol can be replaced. The complete content of the inspection point list can be filtered. The following information is part of the inspection point list:

- AFO - description of characteristics

Both are separated with a hyphen and are shown in the 1. line of a characteristics. Example of a display format: "10" - total length („10" is equivalent to the operation sequence number/AFO).

- Characteristics type

Collection types to be used are "attributive" and "variable".

On the right hand side detailed information (i.e. test equipment and specifications) about the characteristics in the list is shown. The parallel display of the left and right display area is dependent on the selected device. Only the characteristics list is displayed on the smartphone. In order to show details the required characteristic should be selected.

The detailed display enables using special function buttons for the following create/process function:

- Check variable / attributive characteristic (depending on the type of the previously selected characteristics)
- Inspection documents
- List of measured values

5 Inspect Variable Characteristics

The main role of the dialog is collecting measured values from previously selected characteristics. Added information is tolerance limits, measurement unit, target value, sample size and the used test equipment or specified test equipment group.

If the 1. measured value is collected for a characteristics and the sample size is for example 3, the field "Value no." is pre allocated with the "1". If the valued measure and optionally a comment is saved, the collection dialog is automatically closed and opens instantly after storage. The field "Value no" is now pre allocated with " ". If the inspection data could is saved successfully, a message line containing relevant information is sent right after closing the dialog and before the collection catalog is opened again. The message line is located above the characteristics list. If the data is saved the message line is highlighted in green. If it is not saved the color is pink. This can happen if the measured value has not been saved due to a violation of plausibility.

The value no. always increased by one number if the collection dialog is opened again. It might be the case that the value no. "4" is preset but the sample size has the number "3". If no more measured values are to be collected or the specified sample size has been reached, the collection dialog must be closed manually.

6 Inspect Attributive Characteristics

The main role of the dialog is the evaluation from previously selected characteristic. Evaluation is done using information of the number of nonconforming units (field "No of nonconforming units"). Also the number of inspected parts must be collected (inspected sample size). The sample size to be inspected is displayed for information purposes. The inspection result incl. of optional comments are automatically assigned to "Value no" 1.

If the inspection result could be saved successfully, a message line containing relevant information is sent right after closing the dialog and before the collection catalog is opened again. The message line is located above the characteristics list. If the data is saved the message line is highlighted in green. If it is not saved the color is pink. That is the case if there is no network connection.

7 Display Inspection Documents

The overview of the documents can be called from the list of characteristics or the collection dialog for attributive and variable characteristics.

In the document overview a list of documents for the previous selected characteristic or a presently inspected characteristic appears on the left hand side.

The content can be restricted using a filter. A match code filter is available for the user. Content to be filtered in a text is replaced by the symbol "*". Using the symbol "?" an individual symbol can be replaced. The complete content of the document list can be filtered. The following information is part of the document list:

- Document type
Failure type number
The information is displayed in the 1. line of the document entry.
- Description
The information is displayed in the 2. line of the document entry.

In the right display area detailed information is shown in the list for the selected document entry. The parallel display of the left and right display area is dependent on the selected device. Only the document list is displayed on the smartphone. In order to show details the required document entry should be selected.

The fields "Externally" and "Position" are not available on the smartphone.

Document content can be displayed selecting a function button. The content of the field "Text" is shown with the document type "Text". For the type "URL" or "File" the "linked" document is opened with the respective program and displayed. In order to display the URL entry a "Link" in form of <http://www.mpdv.de> is required. An entry in the form of www.mpdv.de cannot be displayed.

8 Display Measured Values / Inspection Results

The measured/single values list derives from the list of characteristics.

On the left hand side this list appears with all previously collected measured values or attributive results.

The content can be restricted using a filter. A match code filter is available for the user. Content to be filtered in a text is replaced by the symbol "*". Using the symbol "?" an individual symbol can be replaced.

The complete content of the document list can be filtered. The following information is part of the measured value list:

- Variable characteristics: Upper tolerance limit, actual value and lower tolerance limit. The information is separated by a pipe slash "|".
This information is displayed in the 1. line.
- Collection point in time (Date and hour)
This information is displayed in the 2. line of the document entry.

In the right display area detailed information is shown in the list for the selected entry. The parallel display of the left and right display area is dependent on the selected device. Only the measured/single values list is displayed on the smartphone. In order to show details the required entry should be selected. In the detailed display also inspection requirements, inspection step number and the corresponding area (i.e. P for production) are shown.